Asking Is (Almost) All You Need: Sequential Reciprocal Matching under Uncertainty

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Code, results and reproduction commands: https://github.com/krishnavallabha/BRVM

Abstract

Each simulated day, a matchmaking policy must decide which clarifying questions to ask under a fixed budget and which feasible, non-overlapping introductions to make, in a changing pool with incomplete profiles and delayed, selective feedback. The objective is *mutual second-meeting intention* (MSMI) per 100 arrived members, and a single invalid action in any of 120 private episodes disqualifies an entry. Our Round 1 proposal, Bayesian Reservation-Value Matching, assumed that lookahead, exploration and rich outcome models would carry the score. In Round 2 we *measured before modelling*.

- We built an exact expected-MSMI yardstick from the public simulator's mechanics, validated it against realised outcomes on 360 episodes, and cut paired evaluation noise about 10×.
- The yardstick shows that selection headroom is small: a matcher that knows every true probability gains only about +7% over greedy. Clarification is the dominant lever: without questions the score falls by 71%.

Our shipped policy combines partner-count question ordering, cheap goal and pace questions, and maximum-cardinality matching whose tie-breaks use observed soft-field fit, a hierarchical empirical-Bayes model of each person's responsiveness and acceptance, and an online Bayesian fit model. On fresh, reserved seeds it improves expected MSMI over the supplied greedy baseline by +0.0115 [+0.0044, +0.0186] per 100 members (+3.0%), with no scenario family clearly worse. It is valid in every episode we ran, including the official container protocol (median 0.95 s per call).

We also report negative results. Waiting for better matches, whether by simple reservation or by our own sampled-future prices, never helps. The models are calibrated but weakly informative. And in sparse graphs the choice among equally large matchings measurably affects future availability.

1 Introduction

Reciprocal recommendation differs from ordinary recommendation in three ways:

- both sides must accept;
- every introduction occupies both people for days;
- outcomes arrive late and only for the pairs one chose to introduce.

The Sequential Matching Problem turns this into a 60-day sequential decision task. Each day has two phases: an *ask* phase, which spends a 12-unit clarification budget on hard-constraint bundles (3 units) or single soft fields (1 unit), and a *match* phase, which proposes a set of disjoint, reciprocally feasible pairs. Scoring is purely technical: the equal-weight mean over six scenario families of MSMI per 100 arrived members, with ties broken by coverage, mutual acceptances, lower clarification cost and lower inference time.

Our Round 1 proposal, Bayesian Reservation-Value Matching (BRVM), committed to a sophisticated design:

- a staged hierarchical outcome model;
- reservation values estimated as LP duals over sampled futures;
- Thompson sampling;
- value-of-information clarification;
- a scenario-aware meta-policy.

This report documents what happened when that design met measurement. Our contributions are:

1. **A validated low-noise yardstick.** Exact expected MSMI per introduction, computed from simulator truth *for evaluation only*. It agrees with realised outcomes (ratio 0.91, $z = -1.53$, 360 episodes) and cuts the paired standard deviation from 0.44 to 0.04 per episode (Section 3).
2. **A headroom analysis** that bounds what any allocation rule can gain and locates the remaining value (Section 3.1).
3. **A simple, robust, confirmed policy.** Each component is ablated, and the policy is confirmed twice on reserved seeds: $+0.0092$ and then $+0.0115$ over greedy (Sections 4–6).
4. **An observable-only Bayesian layer.** A hierarchical empirical-Bayes person model and an online fit model, with a calibration study. They add $+0.0041$ $[+0.0012, +0.0071]$ on fresh seeds and adapt to shifted outcome weights without recognising the scenario (Section 4.1).
5. **Honest answers to the reviewers' questions**, including negative results for our own Round 1 ideas (Section 7).

2 Problem setting

A pool of $n = 200$ synthetic members arrives over days 0–20; about 70% arrive on day 0 and about 12% exit at a random day between 22 and 60.

Fields. Each member has public attributes (age, gender, zone), hard fields $\mathscr{H}$ (age range, genders sought, relationship structure, smoking, children, zones, schedule) and soft fields $\mathscr{S}$ (goal, pace, lifestyle, conversation style, and others). Each field is *observed*, *not asked* or *declined*.

Feasibility. A pair (i, j) is *feasible* iff every hard field of both members is known and every reciprocal check passes. Unknown never means compatible, and a declined hard field makes a member permanently ineligible. On day t the policy may introduce any matching M_t on the graph of feasible, available, never-introduced pairs. After an introduction both members are busy for 8 days, and longer after a mutual acceptance.

Outcomes. For each introduction the simulator draws, in order:

- a response from each member (rate $r_i \in [0.55, 0.98]$);
- acceptance, which depends on a person-level bias, soft-field fit and a shared pair shock;
- a date within 30 days ($p = 0.78$, later in the *delayed* family);
- two second-meeting answers within 3 days.

An introduction counts as an MSMI when all of these succeed. For an episode with N members arrived by day 59, the score is $100\#\text{MSMI}/N$, and the primary metric averages over 20 private seeds within each of six families (*standard, sparse geography, cold start, delayed dates, shifted outcome weights, changing response conditions*), then over families.

Observations. The policy sees only the request JSON: current fields and statuses, availability, past introductions and observed feedback. It never sees r_i, biases or seeds.

3 Measuring before modelling

Expected-MSMI yardstick. Realised MSMI is extremely noisy: about 0.7% of introductions succeed, and an episode has about 1–2 successes. Because the simulator is public, we can compute

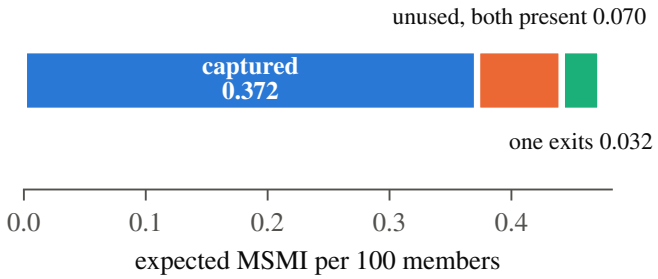


Figure 1: Where the value of feasible pairs goes under `full` (seeds 201–215, all families). Most feasible value is captured. The largest recoverable pool is pairs whose members were both present but never scheduled.

each introduction's exact success probability for *evaluation* (never in the policy):

$$p_{ij} \;=\; r_i^2 r_j^2 \cdot 0.78\,\pi_{30} \cdot 0.36\,\mathbb{E}_\varepsilon\big[\sigma(\ell_{ij}+\varepsilon)\,\sigma(\ell_{ji}+\varepsilon)\,\sigma(s_i+\varepsilon+g_{ij})\,\sigma(s_j+\varepsilon+g_{ij})\big], \qquad (1)$$

where:

- ℓ_{ij} is i's acceptance logit for j;
- s_i is i's second-meeting logit, with g_{ij} its goal-match boost;
- $\varepsilon \sim \mathcal{N}(0, 0.45^2)$ is the shared pair shock, integrated by 40-point Gauss–Hermite quadrature;
- π_{30} is the probability that the date falls inside 30 days;
- $0.36 = (3/5)^2$ is the probability that both second-meeting answers are on time.

The response rate appears squared per person, because each member must answer both the introduction and the second-meeting question. Summing p_{ij} over a policy's introductions gives its expected MSMI.

Validation. On 360 greedy episodes (seeds 201–260, all families) expected and realised totals agree: 265.7 vs 241 events (ratio 0.91, $z = -1.53$), and every family's 95% CI on the per-episode difference contains 0.

Variance reduction. The paired difference between two policies on identical worlds has a per-episode SD of 0.44 when measured by realised MSMI, but 0.041 when measured by expected MSMI. That is a 10× reduction, or about 100× fewer seeds for the same power. Every comparison in this report is paired and uses expected MSMI; realised values are reported alongside.

3.1 How much is there to gain?

On seeds 201–205 the expected MSMI per 100 members is 0.348 for greedy and 0.362 for random feasible selection. A greedy matcher that knows every true p_{ij} reaches 0.360, and a maximum-weight matching on the true p_{ij} reaches 0.372. **A perfect-information myopic allocator gains only about +7%.** Removing all questions loses about 70%.

The reason is structural. The feasible graph is tiny (about 2.5 partners per member), and about 34% of members decline a hard field, which caps coverage at about 0.36.

Figure 1 decomposes the value of all feasible pairs under our policy. Of the value left on the table, about 0.07 per 100 members comes from pairs whose members were both present but never scheduled. Drift timing losses are negligible (0.004). Across feasible pairs, the variance of $\log p_{ij}$ is explained 24% by response rates, 27% by acceptance and second-meeting tendencies, and 36% by goal match (11% in the shifted-weights family). *The signal is mostly about people, not pairs.*

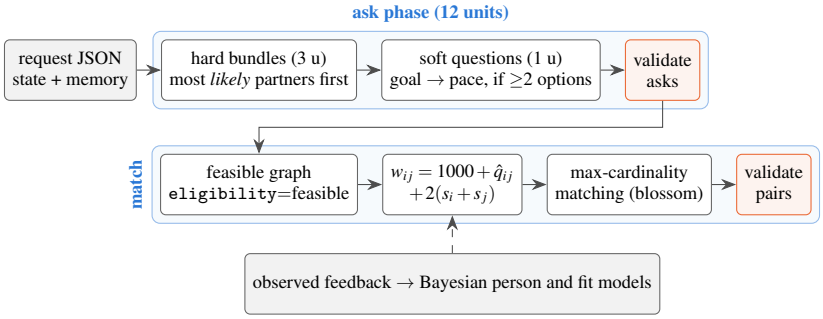


Figure 2: One simulated day. Asks are chosen, validated and resolved. The match phase then builds the feasible graph and solves a maximum-cardinality weighted matching. The weight keeps cardinality first, so models only break ties. Every output passes an independent validator.

4 Method

Figure 2 shows the daily loop. Normalisation, modelling and allocation are kept separate, as the specification recommends. Every component is a configuration switch, so the supplied baselines, our policy and every ablation run through the same code.

Safety shell. Validity is all-or-nothing, so the shell was built and tested before any model. It:

- re-checks every ask (available member, legal field, never observed or declined, no duplicates, budget) and every pair (`kit.eligibility`, availability, no repeats, nobody twice), and drops failures;
- runs the policy under a 3 s soft limit;
- makes the match fallback deadline-aware: past 4.5 s it returns no pairs instead of recomputing greedy;
- checks memory for size and finiteness.

A 36-test suite fuzzes empty pools, all-unknown and all-declined fields, exhausted graphs, malformed memory, and choosers that raise, return garbage or hang. Disabling the validators makes 30 sub-tests fail.

Clarification. Hard bundles go to candidates (available, a hard field not asked, none declined) ranked by *likely partner count*: the number of other available members not yet introduced whose pair status is anything but *infeasible*. Unknown fields therefore do not block. Arrivals in the last 3 days get a $+10$ bonus, tuned over $\{0, 10, 25, 50\}$ on development seeds. Left-over budget asks `relationship_goal` and then `relationship_pace`, only for members with at least two feasible partners, since a soft answer can only change a choice when there is more than one option.

Allocation. Each day we solve

$$M_t = \arg\max_{M \in \mathcal{M}(G_t)} \sum_{(i,j) \in M} \big(1000 + \hat{q}_{ij} + 2\,(s_i + s_j)\big), \tag{2}$$

where $\mathcal{M}(G_t)$ is the set of matchings of the feasible graph G_t, solved with Edmonds' blossom algorithm [1] (maximum-cardinality, maximum-weight). The base weight of 1000 guarantees that cardinality is never traded for quality.

The blossom routine is vendored verbatim from NetworkX [2], BSD-3. Importing the full library cost about 0.93 s on every call, because the policy process restarts per call. The vendored copy is identical on 6,000 random graphs and on full policy episodes.

4.1 Observable-only Bayesian layer

Person model. For member i, let (a_i, n_i) count answered vs total *resolved* response opportunities, from introductions and second-meeting questions, and (y_i, m_i) count yes answers among introduction responses.

Only opportunities whose answer window has closed are counted. A response appears 1–7 days after an introduction, but a non-response only at the 7-day deadline, so counting open opportunities inflated every response rate (calibration showed 0.9 predicted vs 0.7 observed).

We place Beta priors centred on the pool rate $\bar{r}$ with strength κ, and *average* over κ instead of picking one (hierarchical empirical Bayes [3, 4]):

$$\mathbb{E}[f(r_i) \mid \text{data}] \;=\; \sum_{\kappa \in K} w_\kappa \, \mathbb{E}_{\text{Beta}(\bar{r}\kappa + a_i,\, (1-\bar{r})\kappa + n_i - a_i)}[f], \qquad w_\kappa \propto \prod_k \frac{B(\bar{r}\kappa + a_k, (1-\bar{r})\kappa + n_k - a_k)}{B(\bar{r}\kappa, (1-\bar{r})\kappa)}, \tag{3}$$

over a log-spaced grid $K = \{2, \ldots, 400\}$, with w_κ the beta-binomial marginal likelihood. With about two observations per member that likelihood is flat, and a single best κ jumped between 8 and 400 from episode to episode; averaging stabilises the shrinkage. The person score is

$$s_i = \log \mathbb{E}[r_i^2] + \log \mathbb{E}[\text{acc}_i]$$

relative to a member with no data. The square mirrors the outcome structure of Section 3.

Fit model. A Bayesian logistic regression of a yes answer on soft-field agreement features ($+1$ equal, $-\frac{1}{2}$ different, 0 unknown), fitted by MAP with Newton steps. Its prior means are our hand-set weights divided by 4: goal $+3/-1.5$, pace $+1.5/-0.75$, and small lifestyle and conversation terms. With no data, $\hat{q}$ equals the hand-set fit exactly; with data, it adapts.

Sampled-future slot prices (tested, not shipped). Member i's price is

$$\log\Big(1 + \frac{\text{remaining feasible partners} + \text{expected future partners}}{\text{remaining slots}}\Big),$$

where future partners are the recent arrival rate times days left times the share of resolved profiles feasible with i (the sample average over resampled observed profiles), and remaining slots are days left divided by 8. It enters the weight as $\lambda(\text{price}_i + \text{price}_j)$.

5 Experimental protocol

- **Seeds.** Development and all tuning on seeds 201–260. Seeds 301–340 were reserved for confirming `full`, and 341–380 for confirming the Bayesian layer; each set was used once, after the design was frozen. Public seeds 101–103 were used only to reproduce the published baselines.
- **Baselines.** The supplied greedy, no-clarification and random-feasible policies, on identical seeds and all six families.
- **Keep rule, fixed in advance.** A component is kept only if its paired difference against the current best is positive overall, no family's 95% CI lies entirely below zero, and no episode is invalid.

6 Results

Baselines and the official run. Table 1 reports the official protocol. Two observations stand out:

- Our policies produce *more* mutual acceptances and dates than greedy, but at this scale they lost some of the final second-meeting draws.
- In expectation, the shipped policy ties greedy on these three seeds. Three seeds cannot resolve effects of about 1%; the reserved-seed confirmations can.

Table 1: Official evaluator, real subprocess protocol, public seeds 101–103 × 6 families (18 episodes per policy, all valid). Per-episode means. Realised MSMI at this scale rests on 10–14 events per policy; the expected column is far less noisy.

Policy	Realised	Expected	Intros	Mutual acc.	Coverage	Wait med./p90	Ask cost
greedy	0.389	0.373	79.8	11.00	0.372	4 / 16	207
random	0.278	0.370	80.5	10.67	0.377	5 / 16	207
no_asks	0.139	0.116	24.3	3.50	0.143	0 / 13	0
`full` (core)	0.278	0.362	78.5	11.94	0.375	4 / 16	220
`full_person_learn`	0.306	0.373	79.9	11.67	**0.376**	4 / 16	219

Table 2: Paired expected MSMI per 100 members on reserved seeds (240 episodes per policy, all valid), with 95% CIs on the per-seed paired difference.

Seeds	Policy	Expected	vs greedy	vs `full`
301–340	greedy	0.3841	—	
	random	0.3864	+0.0024 [−0.003, +0.008]	
	no_asks	0.1109	−0.2732 [−0.290, −0.257]	
	`full`	0.3933	+**0.0092** [+0.0011, +0.0174]	—
341–380	greedy	0.3779	—	−0.0073
	`full`	0.3852	+0.0073 [+0.0009, +0.0138]	—
	+ person model	0.3888	+0.0109	+0.0036 [+0.0011, +0.0061]
	+ fit model	0.3872	+0.0093	+0.0020 [−0.0003, +0.0042]
	+ both (shipped)	**0.3894**	+**0.0115** [+0.0044, +0.0186]	+**0.0041** [+0.0012, +0.0071]

Confirmation on reserved seeds. Table 2 and Figure 3 give the evidence.

- On 301–340 the core policy `full` beats greedy by +0.0092 [+0.0011, +0.0174].
- On fresh seeds 341–380 the shipped policy beats greedy by +0.0115 [+0.0044, +0.0186], a +3.0% improvement. It is clearly better in the cold-start, delayed and changing-response families, positive in two others, and tied in sparse geography, where about 22 introductions per episode leave nothing to choose between.

Ablations. Figure 4 shows the remove-one ablations.

- The load-bearing design claim is that *allocation must use observed soft-field agreement*. Unit-weight maximum matching loses the whole gain (−0.0082 [−0.0142, −0.0021]).
- Question order, soft questions and maximum cardinality are largely substitutes: any two of them recover about +0.008 over greedy. That is consistent with the low ceiling of Section 3.1.
- Broad soft questioning (every field) matches the narrow design inside `full` (−0.0004) at 2.7× the ask cost, which is a tie-breaker, so the narrow design ships.

The Bayesian layer and calibration. Figure 5 shows reliability diagrams for about 13,000 decision-time predictions.

- All three models are monotone and beat a pool-rate baseline: Brier skill +0.7% for response, +0.55% for acceptance and +0.15% for fit.
- The fit model gains +1.7% skill in the shifted-weights family, where pace outweighs goal. It adapts without being told the family.
- The tails remain overconfident, which is expected with two to three observations per person. The two fixes in Section 4.1 were found through this diagnostic.

A tie-breaking effect. Early on, every new tie-break term lost about 0.01 in the cold-start family, regardless of its sign or strength.

- A *meaningless* deterministic hash tie-break did the same: −0.0063 on seeds 201–230 and −0.0071 on fresh seeds 231–260, about one fewer introduction per episode.

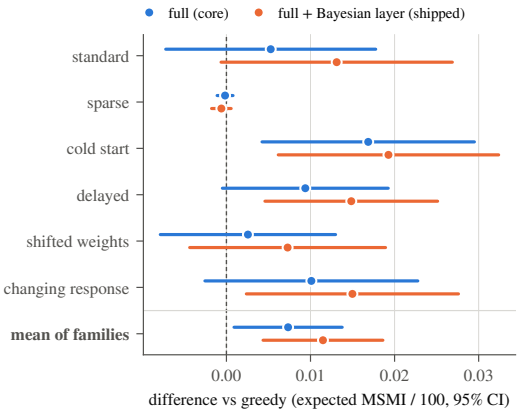


Figure 3: Per-family gains over greedy on fresh seeds 341–380. The Bayesian layer lifts every family except the sparse one, where every policy ties.

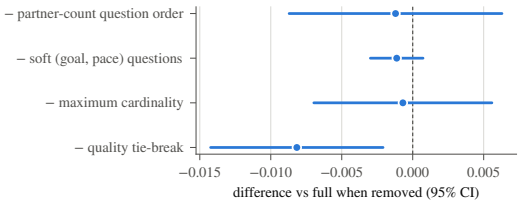


Figure 4: Remove-one ablations of `full` (seeds 301–340). Only removing the soft-fit-aware tie-break is clearly harmful; the other components substitute for each other.

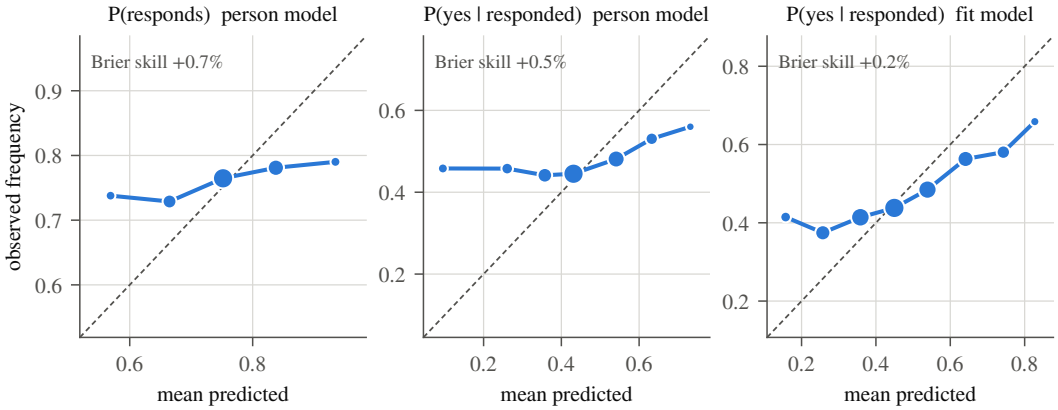


Figure 5: Calibration of the observable-only models at decision time (seeds 201–215, about 13,000 predictions; marker area ∝ bin size).

- So in sparse, mostly unknown graphs, *which* of several equally large matchings is chosen changes future availability. The default blossom order happens to schedule cold starts slightly better.
- With the averaged shrinkage, the shipped layer is positive in cold start (+0.0024 vs `full`).

Validity and runtime. Across thousands of in-process episodes and all official runs we observed no invalid episode.

- An early container run timed out in 2 of 6 episodes. After vendoring the matcher and tightening the time budget, local subprocess calls take 0.46 s on average.
- The final container check is 6/6 valid (median 0.95 s, max 1.07 s per call), with actions identical to the local run.

7 Reviewer questions

Q1: Do sampled-future prices beat zero and simple reservation on unseen arrival mixtures and denser graphs? **No** (Figure 6).

- We built denser stress worlds (2–3 acceptable zones, ages ±12, more multi-gender preferences), with about 3× the introductions.

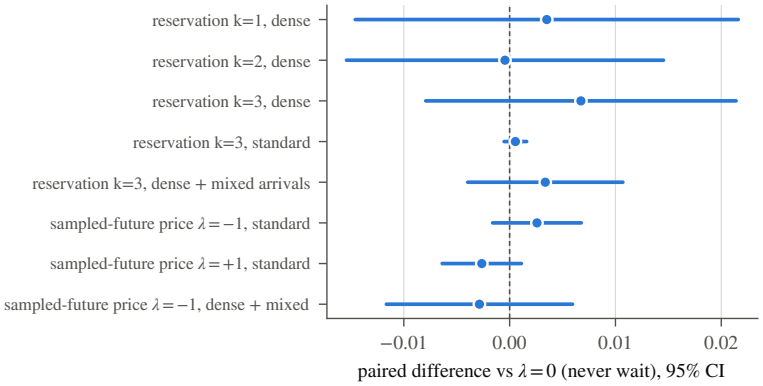


Figure 6: Q1: waiting never helps. Paired differences of simple reservation and sampled-future prices against $\lambda = 0$ (never wait), on standard worlds (seeds 201–230), dense stress worlds and an unseen arrival mixture (seeds 201–260).

- There, simple reservation ($k \in \{1,2,3\}$) and sampled-future prices are never distinguishable from $\lambda = 0$. The same holds on an unseen arrival mixture (40% on day 0, the rest over days 1–40).
- The sign is instructive: keeping scarce members free ($\lambda < 0$, our Round 1 intuition) beats prioritising them ($\lambda > 0$), but neither beats acting now.
- An introduction blocks people for only 8 days, and outcomes are driven more by individual tendencies than by pair fit, so a skipped pair is rarely recovered by a better one.

Q2: Does the policy generalise when conditions overlap or change, without recognising the scenario? Yes. We built world variants by wrapping the generator and subclassing the simulator, without editing the kit. Against greedy (seeds 201–260), `full` scores:

- $+0.0123$ $[+0.004, +0.021]$ when conditions switch on day 30 to drift-style acceptance and delayed dates;
- $+0.0079$ $[-0.002, +0.018]$ with mixed arrivals;
- -0.0006 with sparse and delayed conditions together, where there is nothing to choose between;
- $+0.0065$ $[+0.002, +0.011]$ overall.

We dropped Round 1's scenario classifier (the rules forbid family-specific logic); the online fit model provides continuous adaptation instead.

Q3a: Can complementary questions defeat an independent-item planner? Yes, in principle; ours avoids the trap.

- In 90 initial pools we found 953 truly compatible pairs blocked by *both* members' missing hard fields. Asking either member alone makes 0 of them feasible, so an independent-item planner values each half at exactly zero.
- Such pairs are 22% of all feasible pairs, and 43% in cold start.
- Our partner-count score treats unknown fields as possibly compatible, so both halves score positively. Every such pair had both members asked, and the median wait from co-presence to unlock fell from 6 to 4 days compared with list order.

Q3b: Do response penalties leave members waiting disproportionately? No. The shipped person model down-weights observed non-responders. On seeds 341–380:

- served shares by true response-rate tercile (low, mid, high) are 33.7%, 34.3% and 33.6%, against 33.5%, 34.2% and 33.4% for greedy;
- members with over 50% observed non-responses wait median/p90 $5/17$ days, against $6/20$ under greedy.

Table 3: The Round 1 BRVM hypotheses, judged by Round 2 evidence.

	Round 1 hypothesis	Round 2 verdict
H1	Global max-weight matching beats greedy ranking	**Only with soft-fit weights.** Unit weights -0.005; with quality weights, parity alone and essential inside the full policy
H2	Reservation values (waiting) improve MSMI	**Rejected.** Never distinguishable from $\lambda = 0$, even in dense worlds and unseen arrival mixtures (Q1)
H3	Thompson sampling improves learning	**Not built, deliberately.** Each pair can be introduced once and nearly every feasible edge is used anyway, so exploration has little to change
H4	Value-of-information questions beat random/none	**Supported in spirit.** Cheap proxies (likely-partner count; ask only when there are ≥ 2 options) are cost-efficient; no questions loses 71%
H5	Online updates and scenario adaptation help under change	**Supported without classification.** The online fit adapts in shifted worlds (+1.7% skill), and the policy keeps its gain when conditions switch (Q2)
H6	Correlated reciprocal acceptance improves ranking	**Not tested at policy level.** The shared shock is modelled exactly in our yardstick
H7	A scarcity bonus raises coverage cheaply	**Rejected.** Slot prices change neither coverage (0.325 vs 0.324) nor MSMI

The penalty only breaks ties between equally large matchings: it reorders who goes first, but never removes anyone from the graph.

8 Discussion: Round 1 hypotheses meet the evidence

Table 3 grades our Round 1 proposal against the evidence.

- The parts that survived are the *simple* parts of the proposal: reciprocal feasibility as a hard constraint, global matching, value-of-information-style questioning, and Bayesian shrinkage of person-level effects.
- The parts that were cut (lookahead prices, exploration, scenario classification) all address headroom that the measurements show is not there.

The broader lesson for sequential reciprocal matching is that a validated, low-variance evaluator should come before any modelling ambition. Without it, every effect in this report would be invisible against noise roughly ten times larger.

9 Limitations and broader impact

- **The worlds are synthetic.** Every finding (the low headroom, the value of goal and pace questions, the uselessness of waiting, the tie-breaking effect) is a claim about one published simulator with invented coefficients, not about people.
- **The yardstick inherits the public simulator's structure.** Private worlds from a compatible engine may differ.
- **The priors are hand-set.** The hand-set fit prior and the $+10$ arrival bonus are heuristics tuned on development seeds.
- **The ranking is noisy.** At about ± 0.04 per team on 120 episodes, the realised-MSMI ranking cannot reliably separate competent policies whose expected scores differ by about 0.01.
- **Product use needs more.** Any authorised deployment would need real-time constraint checks, priors re-estimated and recalibrated on consented data, fairness review across protected groups, and a human in the loop.

Reproducibility and attribution. The policy is deterministic and uses only the Python standard library; there are no trained assets and no external data. All commands, seeds and machine-readable results are in the repository. Claude Code (Anthropic) assisted with code, experiments and drafting under the team's direction; the team designed the approach, ran every evaluation and is responsible for all conclusions.

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